

Swiggy's Restaurant Recommendation System using Streamlit

Overview:

The objective is to build a recommendation system based on restaurant data provided in a CSV file. The system should recommend restaurants to users based on input features such as city, rating, cost, and cuisine preferences. The application will utilize clustering or similarity measures to generate recommendations and display results in an easy-to-use Streamlit interface. This project will provide

1. **Personalized Recommendations:** Help users discover restaurants based on their preferences.
2. **Improved Customer Experience:** Provide tailored suggestions to enhance decision-making.
3. **Market Insights:** Understand customer preferences and behaviors for targeted marketing.
4. **Operational Efficiency:** Enable businesses to optimize their offerings based on popular preferences.

Skills:

1. Data Pre-processing
2. One-Hot Encoding
3. Clustering (K-Means, Cosine Similarity)
4. Streamlit Application Development
5. Python

Domain: Recommendation Systems and Data Analytics

Approach:

1. The dataset is provided as a CSV file with the following columns: ['id', 'name', 'city', 'rating', 'rating_count', 'cost', 'cuisine', 'lic_no', 'link', 'address', 'menu']
2. **Categorical:** name, city, cuisine
3. **Numerical:** rating, rating_count, cost

1. Data Understanding and Cleaning

1. **Duplicate Removal:** Identified and drop duplicate rows.
2. **Handling Missing Values:** Imputed rows with missing values.
3. Save the cleaned data to a new CSV file (cleaned_data.csv).

2. Data Pre-processing

1. **Encoding:** Applied One-Hot Encoding to categorical features (name, city, cuisine).
 1. Saved the encoder as a Pickle file (encoder.pkl).
2. Ensured all features are numerical after encoding.
3. Created a pre-processed dataset (encoded_data.csv).
4. Ensured the indices of cleaned_data.csv and encoded_data.csv match.

3. Recommendation Methodology

1. **Clustering or Similarity Measures:**
 1. Used K-Means Clustering and Cosine Similarity to identify similar restaurants based on input features.
 2. Used the encoded dataset for computations.
2. **Result Mapping:**
 1. Mapped the recommendation results (indices) back to the non-encoded dataset (cleaned_data.csv).

4. Streamlit Application

An interactive application with the following components:

1. **User Input:** Accept user preferences (e.g., city, cuisine, rating, price, etc).
2. **Recommendation Engine:** Process the input, query the encoded data, and generate recommendations.
3. **Output:** Displayed recommended restaurants using cleaned_data.csv.

Results:

1. Data Pre-processing

1. Cleaned Dataset (cleaned_data.csv):
 1. Categorical and numerical features with missing values and duplicates removed.
2. Encoded Dataset (encoded_data.csv):
 1. Pre-processed numerical dataset with categorical features One-Hot Encoded.
3. Encoder File (encoder.pkl):
 1. Serialized One-Hot Encoder for Streamlit use.

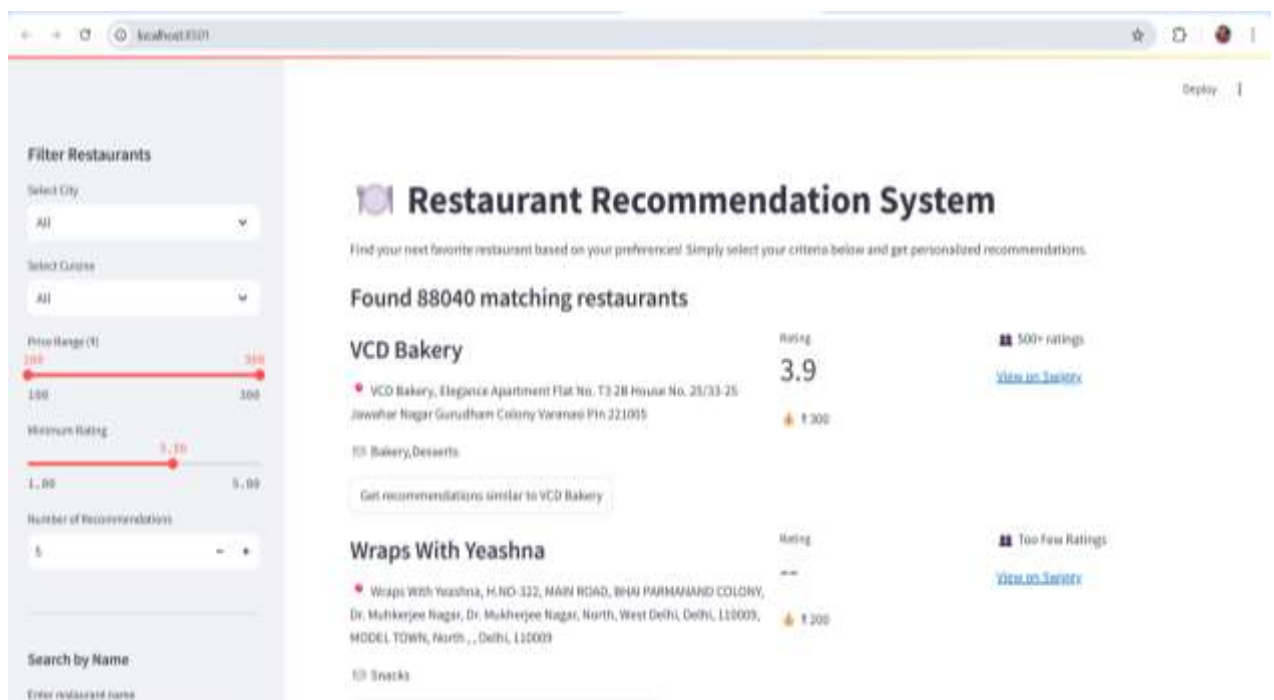
2. Recommendation System

1. Clustering or Similarity-based recommendation engine.
2. Mapping results from encoded_data.csv to cleaned_data.csv for interpretation.

3. Streamlit Application

1. User-friendly interface for input and output.
2. Clear display of recommendations from the cleaned dataset.

Streamlit Screenshots:



← → 🏠 📄 🧑🏻

localhost:8501

☆ 📄 🧑🏻 ⋮

Deploy ⋮

Select City

Adyar,Chennai ▾

Select Cuisine

All ▾

Price Range (₹)

100 250

100 300

Minimum Rating

3.50

3.00 5.00

Number of Recommendations

5 - +

Search by Name

Enter restaurant name

🍛 South Indian,Biryani

Get recommendations similar to ANNA BOJANAM

Vishranthi

📍 Vishranthi, 2, Avenue,TN Housing Board, Besant Nagar, Chennai

🍛 Indian,South Indian

Get recommendations similar to Vishranthi

Pabrai's Fresh and Naturelle Ice Creams

📍 Pabrai's Fresh and Naturelle Ice Creams, Shop # E no 63 6th Avenues Besant Nagar Beach Chennai - 600090

🍦 Ice Cream

Get recommendations similar to Pabrai's Fresh and Naturelle Ice Creams

Rating

4.6

👤 1K+ ratings

[View on Swiggy](#)

👤 20+ ratings

[View on Swiggy](#)

₹ 250

₹ 200

Built with ❤️ by Gokul